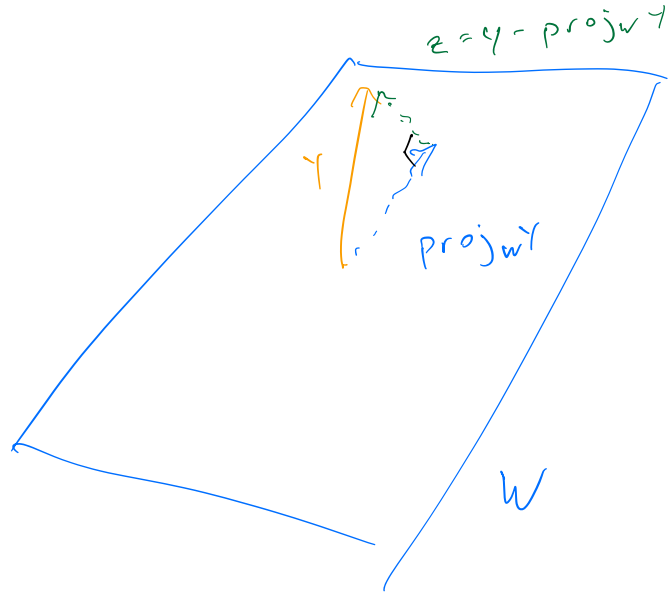

Theorem 8

Let W be a subspace of $\mathbb{R}^n$ (or $\mathbb{C}^n$). Then every vector $y \in \mathbb{R}^n$ (or $\mathbb{C}^n$) may be written <u>uniquely</u> as

$$y = \text{proj}_W y + z$$

where $\text{proj}_W y \in W$ and $z \in W^\perp$ (orthogonal to W).

If $\{w_1, \dots, w_k\}$ is an orthogonal basis for W then

$$\text{proj}_W y = \sum_{l=1}^{k} \frac{\langle y, w_j \rangle}{\langle w_j, w_j \rangle} w_j .$$

Remarks

- In some textbooks Thm 8 is written as

$$\mathbb{R}^n = W \oplus W^\perp .$$

- It follows from Thm 8 that a basis for W together with a basis for $W^\perp$ gives us a basis for $\mathbb{R}^n$. Therefore $\dim \mathbb{R}^n = n = \dim W + \dim W^\perp$.

- In mathematical models there are often some constraints. which amount to choosing a subspace W. If $y \in \mathbb{R}^n$ represents some information from the model, then $\text{proj}_W y \in W$ is the portion of this information relevant to the constraints.

- Once you've computed $\text{proj}_W y \in W$ the orthogonal vector z comes almost for free as

$$z = y - \text{proj}_W y \in W^\perp$$

Example

Take $W = \left\{ \begin{bmatrix} a \\ b \\ b \end{bmatrix} : a, b \in \mathbb{R} \right\} \subseteq \mathbb{R}^3$.

Clearly $W = \left\{ a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} : a, b \in \mathbb{R} \right\}$

$$= \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

So W is a subspace with basis $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$.

Let's check that the basis is orthogonal.

$$\left\langle \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\rangle = (1)(0) + (0)(1) + (0)(1) = 0 .$$

We may use Thm 8 on $y = \begin{bmatrix} 3 \\ 2 \\ 6 \end{bmatrix} \in \mathbb{R}^3$

$$\text{proj}_W \begin{bmatrix} 3 \\ 2 \\ 6 \end{bmatrix} = \frac{\left\langle \begin{bmatrix} 3 \\ 2 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\rangle} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \frac{\left\langle \begin{bmatrix} 3 \\ 2 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\rangle} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{3}{1} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \frac{8}{2} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix}$$

Let $z = y - \text{proj}_W y = \begin{bmatrix} 3 \\ 2 \\ 6 \end{bmatrix} - \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 2 \end{bmatrix}$

By Thm 8 $\begin{bmatrix} 3 \\ 2 \\ 6 \end{bmatrix} = \underbrace{\begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix}}_{\in W} + \underbrace{\begin{bmatrix} 0 \\ -2 \\ 2 \end{bmatrix}}_{\in W^\perp}$

Let's check that $z \in W^\perp$. By some exercise enough to check z orthogonal to basis $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

$$\left\langle \begin{bmatrix} 0 \\ -2 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\rangle = 0$$

$$\left\langle \begin{bmatrix} 0 \\ -2 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\rangle = 0 + (-2)(1) + (2)(1) = -2 + 2 = 0$$:).

Example

Q. What if y in Thm 8 already belongs to W?

A. Clearly $y = y + 0$ where $y \in W$ and $0 \in W^\perp$.

By the <u>uniqueness</u> in Thm 8 $y = \text{proj}_W y$.

Conversely if $y = \text{proj}_W y \in W$ then $y \in W$.

We just prove $y = \text{proj}_W y \iff y \in W$.

Recall from §6.2 that when $W = L$ is 1-dimensional then the closest vector to y in W is $\text{proj}_W y$.

The distance between y and W is defined to be

$\| y - \text{proj}_W y \|$. The same is true for W of higher dimension

Theorem 9 The Best Approximation Theorem

Let W be a subspace of $\mathbb{R}^n$ (or $\mathbb{C}^n$) and y be a vector in $\mathbb{R}^n$ (or $\mathbb{C}^n$). Then the closest point in W to y is $\text{proj}_W y$, i.e.

$$\| y - w \| \geq \| y - \text{proj}_W y \| \qquad \text{for all } w \in W$$

Sketch of proof

$$y - w = \underbrace{(y - \text{proj}_W y)}_{\in W^\perp} + \underbrace{(\text{proj}_W y - w)}_{\in W}$$

$$\Rightarrow \|y - w\|^2 = \|y - \text{proj}_W y + \text{proj}_W y - w\|^2$$

$$= \|y - \text{proj}_W y\|^2 + \underbrace{\|\text{proj}_W y - w\|^2}_{\geq 0.} \quad \text{(Pythagorean Thm Thm 6.1)}$$

$$\geq \|y - \text{proj}_W y\|^2$$

Take square roots.

Def. For W and y as in Thm 9 $\text{proj}_W y$ is the best approximation of y by W and the distance from y to W is

$$\|y - \text{proj}_W y\|.$$

Example

Let $W = \text{span}\left\{ \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix} \right\} \subseteq \mathbb{C}^3.$

Let's check that the basis $\left\{ \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix} \right\}$ is orthogonal.

$$\left\langle \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix} \right\rangle = (1)\overline{(1)} + (i)\overline{(-i)} + (0)\overline{(2i)}$$

$$= 1 + (i)(i) + 0$$

$$= 1 + -1$$

$$= 0.$$

Take $y = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$. Then as in Thm 8

$$\text{proj}_W y = \frac{\left\langle \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} \right\rangle} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} + \frac{\left\langle \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix}, \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix} \right\rangle} \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix}$$

$$= \frac{(1)(1) + (1)(\overline{i}) + 0}{1^2 + (i)(\overline{i}) + 0} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} + \frac{1 + i + 0}{1^2 + 1^2 + 2^2} \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix}$$

$$= \frac{1-i}{1+1} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} + \frac{1+i}{6} \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix}$$

$$= \frac{1-i}{2} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} + \frac{1+i}{6} \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix}.$$

$$= \cdots$$

$$= \frac{1}{3} \begin{bmatrix} 2-i \\ 2+i \\ -1+i \end{bmatrix}$$

The distance from $y = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ to W is

$$\| y - \text{proj}_W y \| = \left\| \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 2-i \\ 2+i \\ -1+i \end{bmatrix} \right\|$$

$$= \cdots$$

$$= \left\| \frac{1}{3} \begin{bmatrix} 1+i \\ 1-i \\ 1-i \end{bmatrix} \right\| = \frac{1}{3} \left\| \begin{bmatrix} 1+i \\ 1-i \\ 1-i \end{bmatrix} \right\|$$

$$= \frac{1}{3} \sqrt{|1+i|^2 + |1-i|^2 + |1-i|^2}$$

$$= \frac{1}{3}\sqrt{(1^2+1^2)+(1^2+1^2)+(1^2+1^2)}$$

$$= \frac{1}{3}\sqrt{6} = \frac{\sqrt{6}}{3}.$$

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§ 6.4 The Gram-Schmidt Process

Typically when we want to compute $\text{proj}_W y$ the subspace W is only given to us as the span of a basis. To compute $\text{proj}_W y$, we need an <u>orthogonal</u> basis. The Gram-Schmidt process converts a basis to an orthogonal basis.

Example 2 in textbook.

Start with a lin. independent set $\{x_1, x_2, x_3\} \subseteq \mathbb{R}^4$

where $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

The Gram Schmidt process proceeds in steps.

Step 1. Set $v_1 = x_1$

Step 2. Set $v_2 = x_2 - \text{proj}_{v_1} x_2$ (by Thm 8 v_2 is orthogonal to v_1)

$$= x_2 - \frac{\langle x_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{\left\langle \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{3}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Step 3 $\quad v_3 = x_3 - \underbrace{\left(\frac{\langle x_3, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2 + \frac{\langle x_3, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 \right)}_{\text{proj}_W x_3 \text{ where } W = \text{span}\{v_1, v_2\}} \in W^\perp$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{\left\langle \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle}{\left\langle \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle} \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$- \frac{\left\langle \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \left(\frac{\frac{2}{4}}{\frac{12}{8}} \right) \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$